

Road Transparency – AI-Powered Road Damage Detection and Complaint Management System

1. Project Overview

Road Transparency is an AI-driven road damage monitoring and complaint management platform designed to help citizens report road defects such as potholes and cracks while enabling government authorities to prioritize repairs efficiently.

The system combines Computer Vision, Geolocation, Complaint Management, Analytics, Reward Mechanisms, and Role-Based Authentication to create a transparent and scalable road maintenance ecosystem.

2. Problem Statement

Road damages such as potholes and cracks often remain unreported or unresolved for long periods, leading to:

- Increased road accidents
- Vehicle damage
- Traffic congestion
- Delayed maintenance activities
- Lack of transparency in complaint resolution

The proposed solution automates road damage detection and prioritization using Artificial Intelligence.

3. Dataset Preparation

The project utilizes the Road Damage Detection Dataset (RDD) obtained from Kaggle.

Dataset Processing Steps

1. Download RDD dataset.
2. Extract only India road images.
3. Extract corresponding annotation files.
4. Filter images containing:
5. Potholes

6. Alligator Cracks
 7. Convert annotations into YOLO format.
 8. Split dataset into:
 9. Train Set
 10. Validation Set
 11. Test Set
 12. Create `data.yaml` configuration file.
 13. Prepare dataset for YOLOv8 training.
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4. AI Model Development

Model

YOLOv8n (You Only Look Once Version 8 Nano)

Training Configuration

- Model: YOLOv8n
- Image Size: 640 × 640
- Epochs: 20
- Batch Size: 8
- Patience: 10
- Workers: 4

Target Classes

Class ID	Damage Type
0	Crack
1	Pothole

Model Output

The trained model returns:

- Damage Type
- Confidence Score
- Bounding Box Coordinates

The final trained model is stored as:

```
backend/models/best.pt
```

5. Severity Assessment Engine

After detecting damage, the system calculates severity.

Severity Formula

```
Severity =  
√(Area Ratio)  
× Confidence  
× Damage Weight  
× 100
```

Where:

```
Area Ratio = Bounding Box Area / Image Area
```

Damage Weights

Damage Type	Weight
Crack	0.8
Pothole	1.0

Severity Categories

Score	Category
0 – 25	Low
26 – 50	Medium
51 – 75	High
76 – 100	Critical

6. Duplicate Complaint Detection

To avoid multiple reports for the same road damage:

Duplicate Rules

The system checks:

- Same damage type
- Within 50 meters
- Reported within last 7 days

Technique

Haversine Distance Formula

This reduces redundant complaints and improves database efficiency.

7. Priority Prediction Engine

The system calculates repair priority automatically.

Priority Formula

```
Priority =  
(Severity × 0.6)  
+ Report Score  
+ Road Importance  
+ Pending Score
```

Road Importance Weights

Road Type	Weight
School Zone	20
Hospital Zone	20
Main Road	15
Bus Route	15
Residential Area	10

Priority Categories

Score	Category
0 – 30	Low

Score	Category
31 – 60	Medium
61 – 80	High
81 – 100	Critical

This enables authorities to focus on the most dangerous road defects first.

8. Backend Architecture

Framework

FastAPI

Advantages

- High performance
 - Automatic Swagger API documentation
 - Asynchronous support
 - Easy frontend integration
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9. Database Design

SQLite is used as the primary database.

Tables

Users

Stores:

- User ID
- Name
- Email
- Role
- Points
- Creation Date

Complaints

Stores:

- Complaint ID
- User ID
- Image Path
- Latitude
- Longitude
- Damage Type
- Confidence
- Severity
- Priority
- Status

Repair Updates

Stores:

- Repair Image
- Remarks
- Update Date

Rewards

Stores:

- User Points
 - Badge Information
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10. Authentication System

Firebase Authentication is used.

Features

- User Registration
- Login
- Token Verification
- Secure Authentication
- Role-Based Access Control

Roles

Citizen

Can:

- Report road damage
- Track complaints
- View rewards

Authority

Can:

- Review complaints
- Update repair status
- Upload repair images

Admin

Can:

- Manage users
 - Monitor analytics
 - Access full dashboard
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11. Complaint Reporting Workflow

Step 1

Citizen uploads:

- Road image
- Latitude
- Longitude

Step 2

Backend performs:

- Image storage
- YOLO detection
- Severity calculation
- Duplicate detection
- Priority calculation

Step 3

Complaint is stored in database.

Step 4

Authorities receive complaint for action.

12. Repair Monitoring System

Authorities can:

- Update complaint status
- Upload repair images
- Add repair notes

Complaint Lifecycle

```
Pending
  ↓
Under Review
  ↓
Assigned
  ↓
In Progress
  ↓
Resolved
```

Citizens can track every stage.

13. Reward System

To encourage citizen participation:

Points

Action	Points
Verified Complaint	10
Resolved Complaint	20

Badges

Points	Badge
< 50	Bronze
< 150	Silver
< 300	Gold
300+	Road Guardian

Leaderboard

Ranks users based on total contribution points.

14. Geographic Visualization

The system generates interactive maps using Folium.

Features

Damage Map

Displays complaint locations.

Heatmap

Shows concentration of road damage reports.

Color Coding

Severity	Color
Low	Green
Medium	Yellow
High	Orange
Critical	Red

This helps authorities identify high-risk regions.

15. Analytics Dashboard

The dashboard provides:

Complaint Statistics

- Total Complaints
- Pending Complaints
- Resolved Complaints
- Critical Complaints

AI Insights

- Damage Type Distribution
- Severity Distribution
- Priority Distribution
- Geographic Distribution

Administrative Reports

- Resolution Rate
 - Citizen Participation
 - Repair Performance Metrics
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16. API Modules

Authentication APIs

- Register
- Login
- Profile

Complaint APIs

- Report Complaint
- View Complaints
- View Complaint Details
- Delete Complaint
- Update Status

Repair APIs

- Upload Repair Evidence
- Add Repair Notes

Dashboard APIs

- Statistics
- Critical Cases
- Pending Cases
- Resolved Cases

Map APIs

- Interactive Map
- Heatmap

Analytics APIs

- Data Distribution
- Dashboard Metrics

Rewards APIs

- User Rewards
 - Leaderboard
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17. Deployment Architecture

Backend

FastAPI

AI Model

YOLOv8n

Authentication

Firebase Authentication

Database

SQLite

Mapping

Folium

Frontend

React.js / Next.js

18. Expected Outcomes

The proposed system will:

- Automatically detect road damages using AI.
 - Reduce manual inspection efforts.
 - Prioritize repairs intelligently.
 - Prevent duplicate complaint submissions.
 - Improve transparency in road maintenance.
 - Provide real-time complaint tracking.
 - Encourage citizen participation through rewards.
 - Assist authorities with analytics-driven decisions.
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Conclusion

Road Transparency is a complete AI-powered road maintenance platform that integrates YOLOv8-based damage detection, severity assessment, priority prediction, geospatial visualization, complaint tracking, Firebase authentication, analytics dashboards, and citizen reward mechanisms into a unified system. The solution improves road safety, increases transparency, and enables faster and smarter road repair management.